Sudarsh Kunnavakkam

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Work Experience

CTO, Co-Founder Nov 2025 — Present
Cognitogram Labs *San Francisco, CA*

- Co-founded venture-backed neuroscience startup pursuing whole-brain emulation; raised $1M pre-seed from Character Capital at 18.
- Built an imaging facility from the ground up to image the whole brain of the zebrafish at 10,000 slices per second in 3 months
- Designed and trained autoregressive models of larval zebrafish neural activity toward whole-brain emulation.

Research Assistant Sep 2023 — Jun 2025
Model Evaluation and Threat Research (METR) *Berkeley, CA*

- Lead engineer for internal project to estimate the agentic time horizon of LLMs at much lower cost, through distilling failure points in agentic transcripts
- Co-lead engineer of a evaluation for Chain-of-Thought Faithfulness of Large Language Models, finding that Chain of Thought is informative about LLM cognition as long as the cognition is complex enough that it can't be performed in a single forward pass
- Helped lead teams of contractors red-team LLMs and curate datasets such as DAFT Math of difficult, free-response questions

Undergraduate Research Intern Nov 2024 — Aug 2025
ShapiroLab at Caltech *Pasadena, CA*

- Built a high-throughput ultrasound screening platform; designed custom proteins with RFDiffusion, AlphaFold, and ESM3 for faster ultrasound reporter kinetics.

Research Fellow Feb 2025 — May 2025
Supervised Program for Alignment Research *Remote*

- Implemented a *continuous double auction* agent arena as a model environment for LLM collusion, accepted to *ICML 2025*

Education

California Institute of Technology Pasadena, CA
B.S. in Physics & Computer Science *In progress, on leave*

- Co-director of Caltech AI Alignment, lead reading groups, research, and invited speakers; scaled from 10 to 100+ attendees at events

Writing & Other Work

1. Simulated Qualia Mugging
2. Takes on Automating Alignment
3. How Should We Implement an AI Pause?
4. Redundant Attention Heads in Large Language Models For In Context Learning

Selected Publications

1. A. Deng*, S. Von Arx*, B. Snodin, S. Kunnavakkam, T. Lanham, "CoT May Be Highly Informative Despite "Unfaithfulness"" by *METR*, Aug 2025
2. K. Agarwal, V. Teo, J. Vazquez, S. Kunnavakkam, V. Srikanth, A. Liu, "Evaluating LLM Agent Collusion in Double Auctions" at *ICML 2025 Workshop on Multi-Agent Systems in the Era of Foundation Models* , Vancouver, Canada, July 2025.
3. C. J. Effarah*, T. Chen*, S. Kunnavakkam*, C. M. Gonzalez, H. W. Lee, "Liquid Metal Printed 2D ITO for Nanophotonic Applications," in *California-US Government Workshop on 2D Materials*, Irvine, California, USA, Sep 2023

Projects

METR: Faithfulness and Monitorability Eval 2025

- Co-lead engineer on METR research report on chain-of-thought (CoT) faithfulness (Aug 2025), extending Anthropic's seminal evaluation to three frontier models and publishing findings for the wider safety community

LLM Agent Collusion Arena 2025

- Implemented a continuous double auction system for agents
- Implemented oversight, monitors, and other experimental conditions to test influence on collusion
- Added logging and metrics with WandB
- Accepted to ICML 2025 Workshop on Multi-agent Systems

Scanning Tunneling Microscope 2024

- Built working STM for $1,000 using open-source design
- Achieved atomic-resolution imaging capabilities (Circuit Design, Signal Processing, Mechanical Engineering)

Awards

ARENA 6.0 Attendee 2025

Non-trivial Fellow 2024

YC W26 Acceptance (Declined) 2026